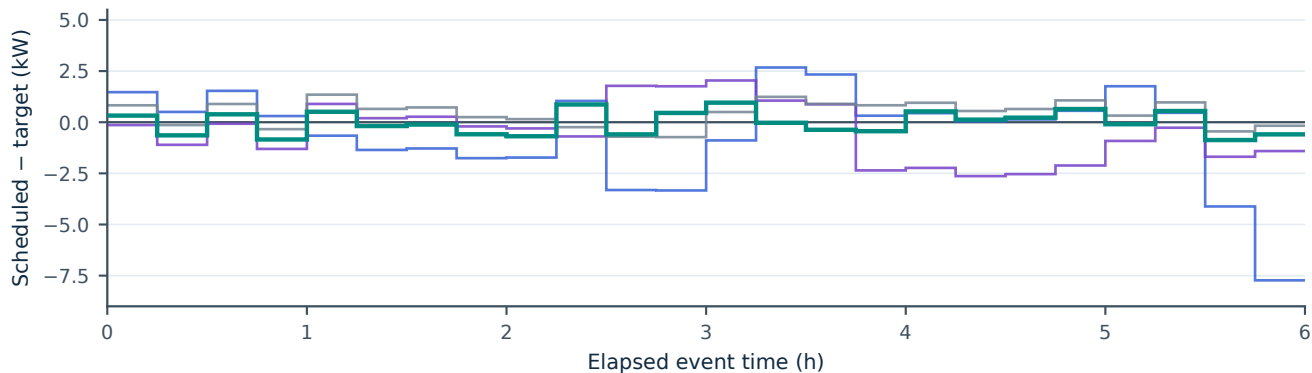
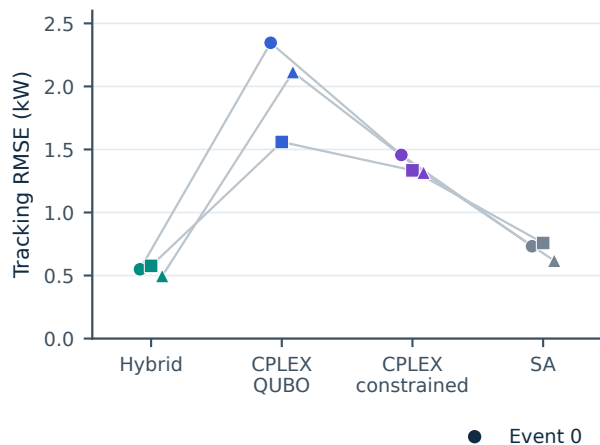


— D-Wave hybrid — CPLEX-QUBO — CPLEX-constrained — SA

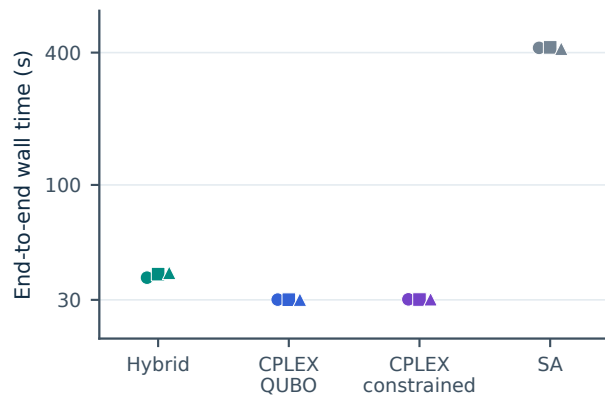
a Event 0: physical tracking differences



b Quality on three benchmark events



c Actual elapsed time



Hybrid and CPLEX: 30 s requested solver limit. SA: 417-422 s wall time.

CPLEX-constrained adds a hard envelope; the other methods use the same QUBO. No optimality or quantum-advantage claim.